

Case 01: Cranial Plate Design

Design and Development of a Patient-Specific Cranial Plate and Mold Using Medical Imaging and Additive Manufacturing Techniques

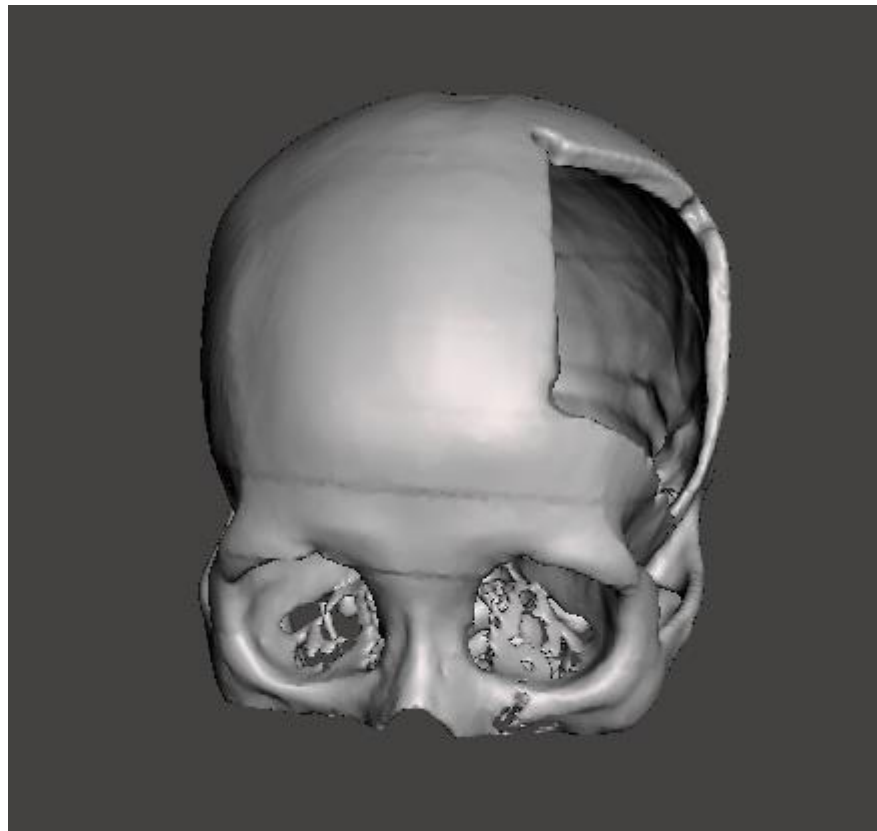
1. Introduction

1.1 Cranial Defect

Cranial defects are discontinuities in the skull caused by trauma, decompressive craniectomy, tumor resection, congenital abnormalities, or infections. These defects can compromise brain protection, aesthetics, and patient quality of life. Restoration of the cranial vault is achieved through a surgical procedure known as cranioplasty. PMMA is commonly used because of its low cost, biocompatibility, and ease of fabrication.

1.2 Need for Patient-Specific Implant (PSI)

Conventional manually shaped implants often result in poor anatomical fit and longer surgical time. Patient-Specific Implants (PSIs) are designed directly from CT images and accurately match the patient's anatomy, leading to improved cosmetic and functional outcomes.



2. Objective

The objective of this case study is:

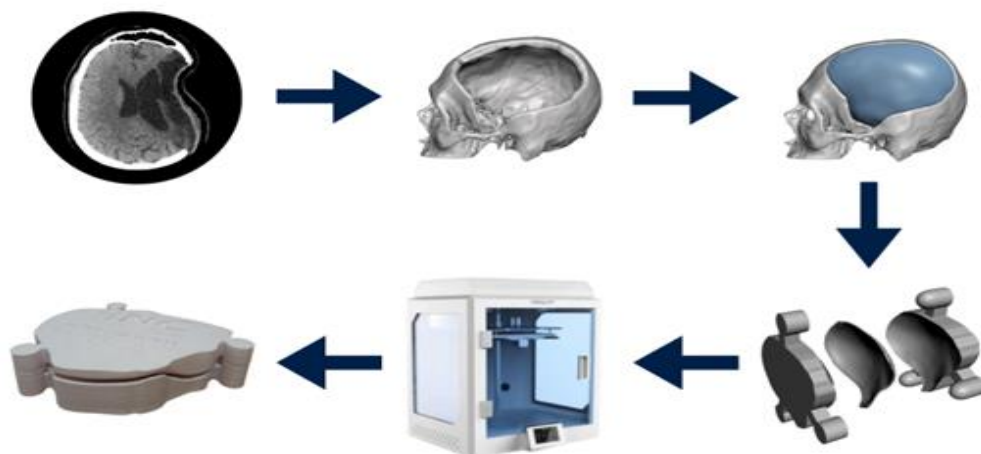
- To design a patient-specific cranial implant from CT data.
- To reconstruct the skull defect using mirror anatomy.
- To generate a mold for PMMA implant fabrication.
- To demonstrate a low-cost additive manufacturing workflow.
- Manufacturing of Titanium and Peek Plate

3. Materials and Software Used

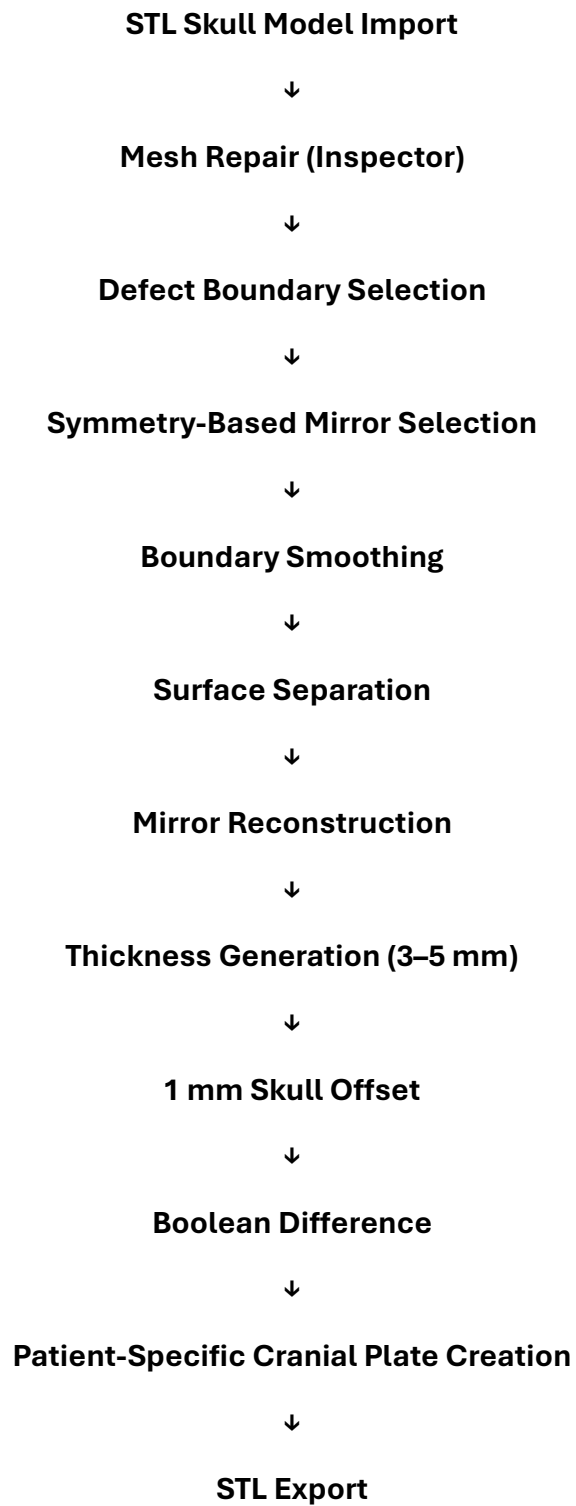
Component	Software / Material
Medical Imaging	CT Scan (DICOM)
Segmentation	D2P / 3D Slicer
3D Reconstruction	D2P
Implant Design	Autodesk Meshmixer
Slicing	Cura
Additive Manufacturing Material	Ti-6Al-4V, PEEK, PLA
Printing Technology	SLM (Titanium), FDM/SLS (PEEK)

Approaches:

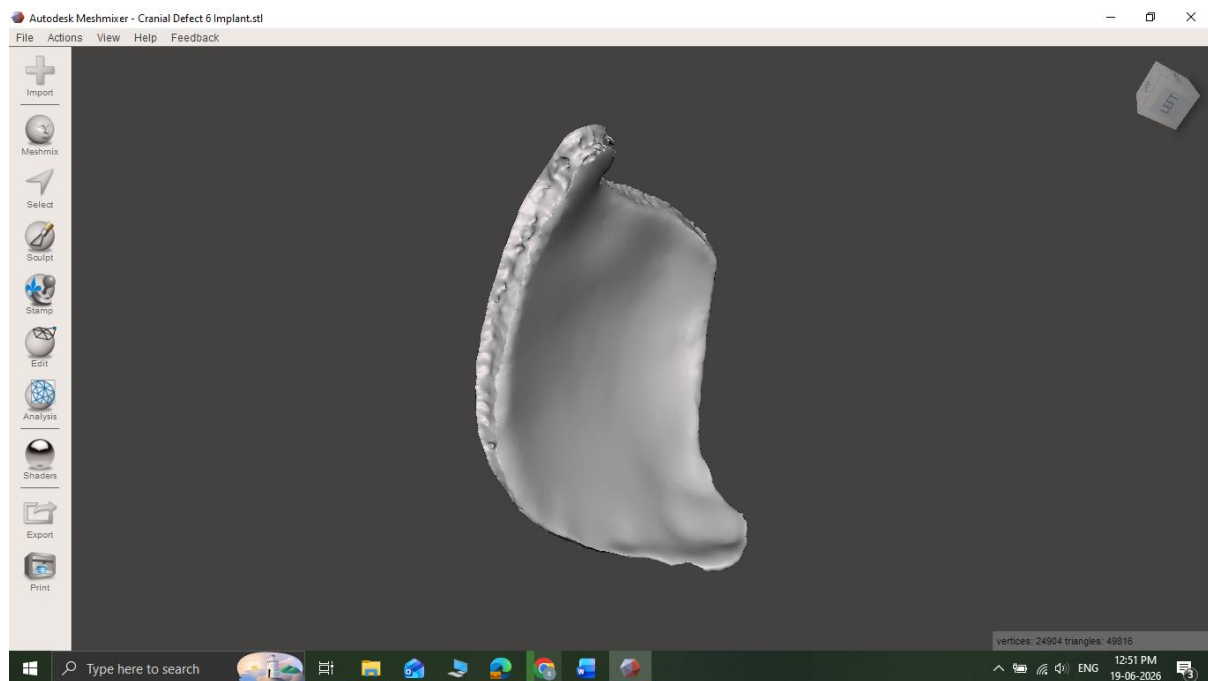
1. Patient-Specific PMMA Cranial Implant Using Mold-Based Manufacturing Approach:



Method:



Result



B) Design Specifications of Titanium PSI Cranial Implant

Parameter	Specification
Implant Type	Patient-Specific Cranial Implant
Design Method	Mirror Reconstruction
Material	Ti-6Al-4V ELI
Implant Thickness	3–5 mm
Central Region	Circular Perforations
Edge Region	Gyroid Lattice
Gyroid Width	20–40 mm (2–4 cm)
Purpose of Gyroid	Bone Ingrowth & Osseointegration
Fixation Method	Titanium Screws
File Format	STL

Design Specifications of PEEK PSI Cranial Implant

Parameter	Specification
Implant Type	Patient-Specific Cranial Implant
Design Method	Mirror Reconstruction
Material	Medical Grade PEEK
Implant Thickness	2.5–4.0 mm
Central Region	Circular Perforations
Edge Region	Gyroid Lattice
Gyroid Width	10–20 mm
Purpose of Gyroid	Tissue Integration & Weight Reduction
Fixation Method	Titanium Cranial Screws
File Format	STL

